MADOCALIB ver.2.2 Manual

QZSS Strategy Office, Cabinet Office

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1 Overview

This manual describes how to use the MADOCA-PPP (Multi-GNSS Advanced Orbit and Clock Augmentation - Precise Point Positioning) [1] test library called MADOCALIB. The purpose of distributing MADOCALIB is to facilitate the user algorithm implementation of MADOCA-PPP. MADOCALIB is constructed based on RTKLIB [2] and functions of PPP-AR and message conversion copyrighted by the third party and provides RNX2RTKP and CSSR2SSR.

RNX2RTKP is a utility for post-process positioning. It implements a processing mode called PPP (Precise Point Positioning) to make use of MADOCALIB. In this processing mode, the following functions are added to RTKLIB ver.2.4.3b34:

- To use the Galileo E1-E5a signal for PPP calculation
- To read the Compact SSR message of MADOCA-PPP
- To select GNSSs to be used for PPP calculation, and
- To select frequencies to be used for PPP calculation for each GNSS
- To process the ionospheric correction data of MADOCA-PPP
- To support triple/quad-frequency PPP

Users can refer to its output as a reference to achieve the performance criteria of MADOCA-PPP.
MADOCALIB also has a processing mode called PPP-AR (Precise Point Positioning-Ambiguity Resolution). Please note that the performance of MADOCA-PPP with the PPP-AR method is currently not defined in the specification documents.

CSSR2SSR is a utility for post-processing conversion of L6E message file containing MADOCA-PPP compact SSR data to RTCM3 SSR message file.

MADOCALIB_GUI is a Windows GUI application that allows users to configure and execute the console-based RNX2RTKP tool and launch RTKPLOT to visualize processing results.

Note: The screenshots in this manual may differ from the actual screens depending on the version of MADOCALIB.

[1] IS-QZSS-MDC-004: https://qzss.go.jp/en/technical/ps-is-qzss/ps-is-qzss.html
[2] RTKLIB: https://github.com/tomojitakasu/RTKLIB.git

2 Getting Start

2.1 System Requirements

The executable binary CUI apps included in the package require Microsoft Windows environment. On the other OS (Linux, etc.) or environment, you have to compile and build CUI apps by yourself. The executable CUI apps are built with mingw-w64, which can be obtained from the following URL. They have been tested on Windows 11 (64-bit).
URL: https://www.mingw-w64.org/

In addition to the CUI apps, this package includes a GUI application implemented in Python.
The GUI app is provided as standalone executable (.exe) files and was built and tested on Microsoft Windows 11 (64-bit) using Python 3.14.2. When not using the provided executable (.exe), the tkinterdnd2 library is required.
Its operation on Linux or other non-Windows environments has not been tested. It is expected that the GUI app should be rebuilt from the provided Python sources if execution in such environments is required.

2.2 Installation

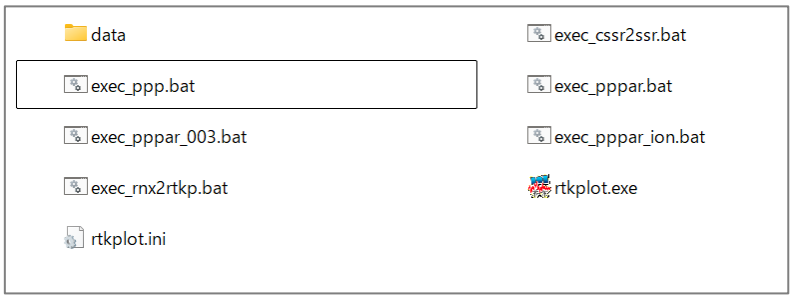
Extract the program package `madocalib` to appropriate directory *<installed_dir>*.on the local disk. The directory structure of the package is as below.

<installed_dir>			Directory where this software is installed
	¥app¥consapp		The application resources are stored.
		¥cssr2ssr	The source files and makefile of CSSR2SSR are stored.
		¥rnx2rtkp	The source files, makefile and sample configuration files of RNX2RTKP are stored.
	¥app¥guiapp	¥madocalib_gui	The source files of MADOCALIB_GUI are stored.
	¥bin		Executable binary of apps for Windows
	¥doc		Document source files
	¥sample_data		Sample data and sample BAT files for apps.
	¥src		Source files of MADOCALIB library.

2.3 Quick Start with Sample Data

Here are quick start guide for RNX2RTKP and CSSR2SSR using the sample data included in this package.

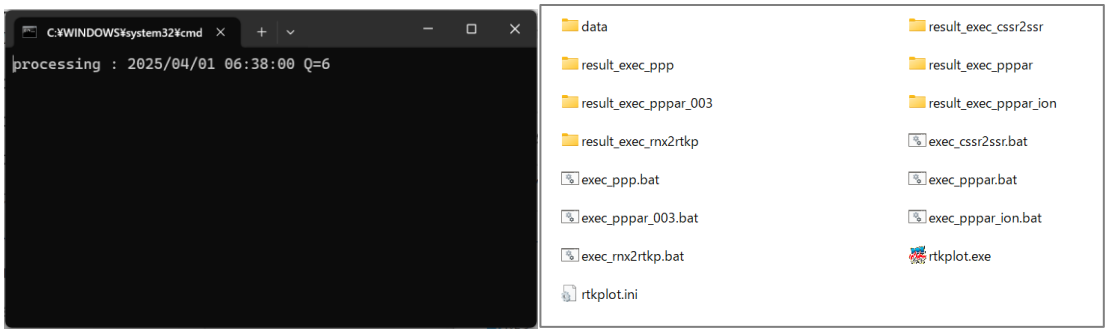
Open the following folder on “Windows Explorer”.
`<installed_dir>¥sample_data`



There are BAT files which execute PPP/PPP-AR from 00min00sec to 59min30sec every hour.

`exec_ppp.bat` : for PPP (no integer ambiguity resolution) by RNX2RTKP
`exec_pppar.bat` : for PPP-AR (with integer ambiguity resolution) by RNX2RTKP
`exec_pppar_ion.bat`: for PPP-AR with ionospheric correction by RNX2RTKP
`exec_cssr2ssr.bat` : for converting cssr to ssr by CSSR2SSR

Double-click on the above BAT file for each scenarios. You can see the command prompt window and get output solution files in the result folder of `result_<BAT filename>`.

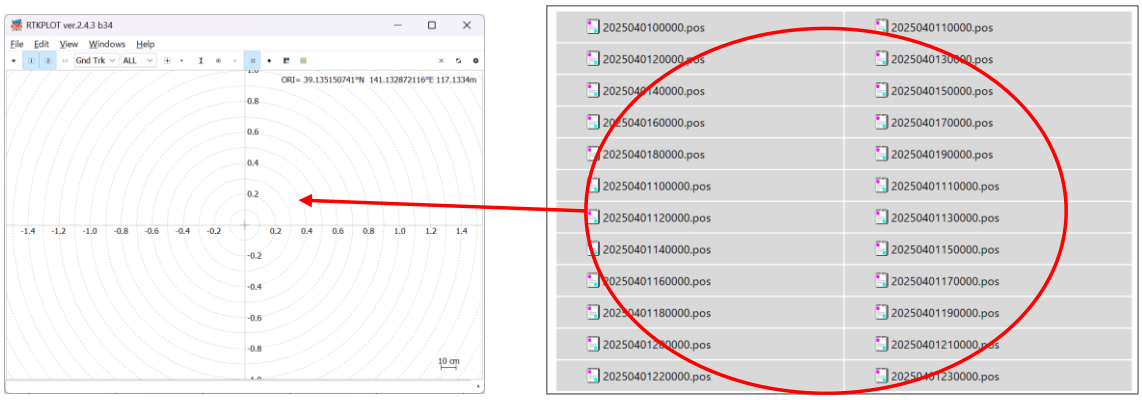


The file output by RNX2RTKP contains the receiver positions calculated by PPP, which can be visualized by RTKPLOT included in RTKLIB package.
The RTKPLOT is included in the folder of `sample_data`, you can use it to visualize the following steps.

- Double-click `rtkplot.exe` to start it.
- Click to select the RNX2RTKP output `*.pos` files in the result folder of `result_<BAT`

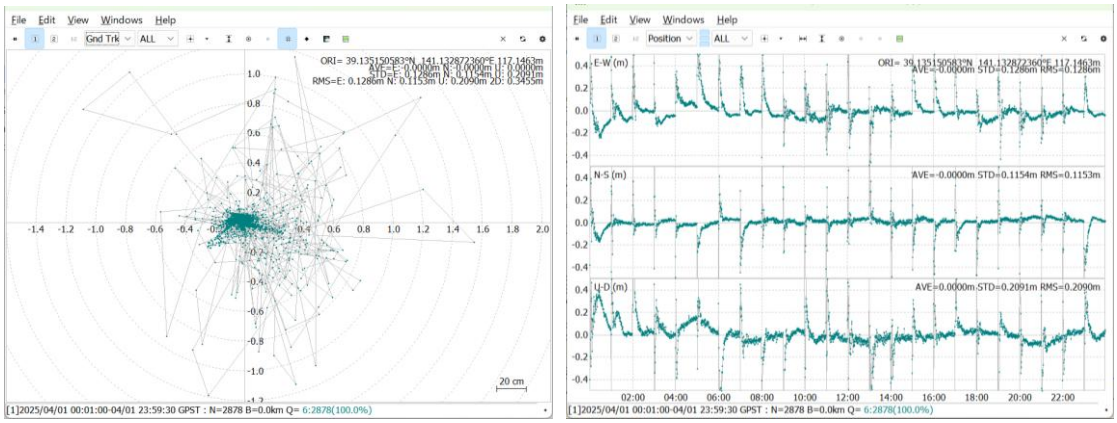
filename> to visualize.

- Drag and drop selected `*.pos` files into the RTKPLOT



A time series plot can be viewed by selecting “Position” from the pull-down menu in the upper left corner of the RTKPLOT. For the detail of RTKPLOT, see RTKLIB manual [4].

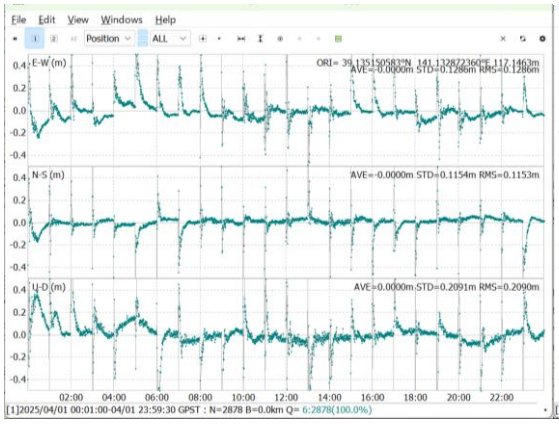
[3] RTKLIB manual: https://github.com/tomojitakasu/RTKLIB/blob/master/doc/manual_2.4.2.pdf



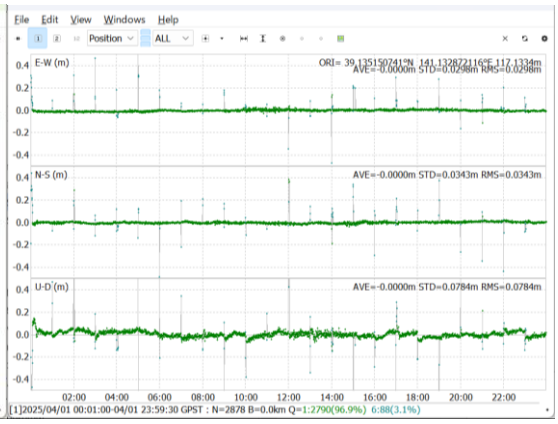
By open each batch file in a text editor, you can see the input data, output data, location of the configuration file, and how to specify the commands used in each.

Sample data corresponding to IS-QZSS-MDC-005 is stored and PPP scenarios are prepared. The sample data are the RINEX files and the L6E messages of PRN 207 on March 28, 2026, in which the PRNs of the BeiDou satellites are rewritten to those after the BeiDou PRN reassignment of April 2026 for testing. The file names of the RINEX files end with _BDSTEST. In addition, sample data corresponding to IS-QZSS-MDC-004 on April 1, 2025, before the reassignment, is stored for exec_pppar_ion.bat.

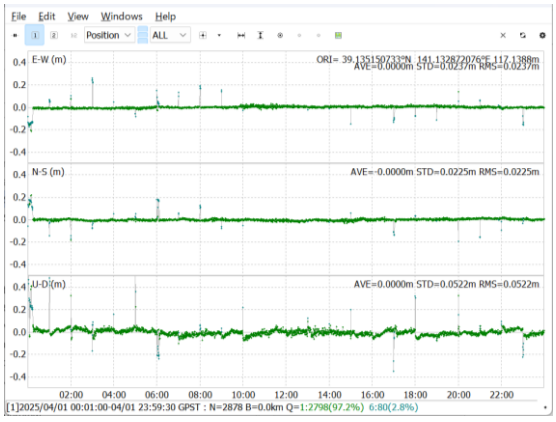
Sample output results for each scenario are shown below.



exec_ppp.bat



exec_pppar.bat



exec_pppar_ion.bat

2.4 Quick Start with Downloaded Data

Here is a quick start guide for RNX2RTKP using the downloaded data.

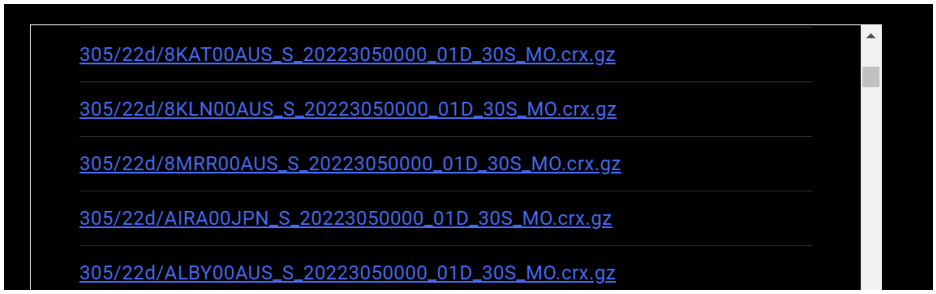
(1) Getting RINEX OBS and NAV files

The RINEX OBS and NAV data can be downloaded from the following IGS's and MIRAI's URL.

IGS<https://igs.org/data/#daily_data>

MIRAI<https://go.gnss.go.jp/mirai/miraiarchive/>





RINEX OBS files are saved in Compact RINEX format, so converting them to RINEX v3 is required. The RNXCMP (a tool that converts the Hatanaka compact RINEX format (extension crx) is used to convert to the commonly used RINEX format (extension rnx)) and can be obtained from the following

https://terras.gsi.go.jp/ja/crx2rnx.html

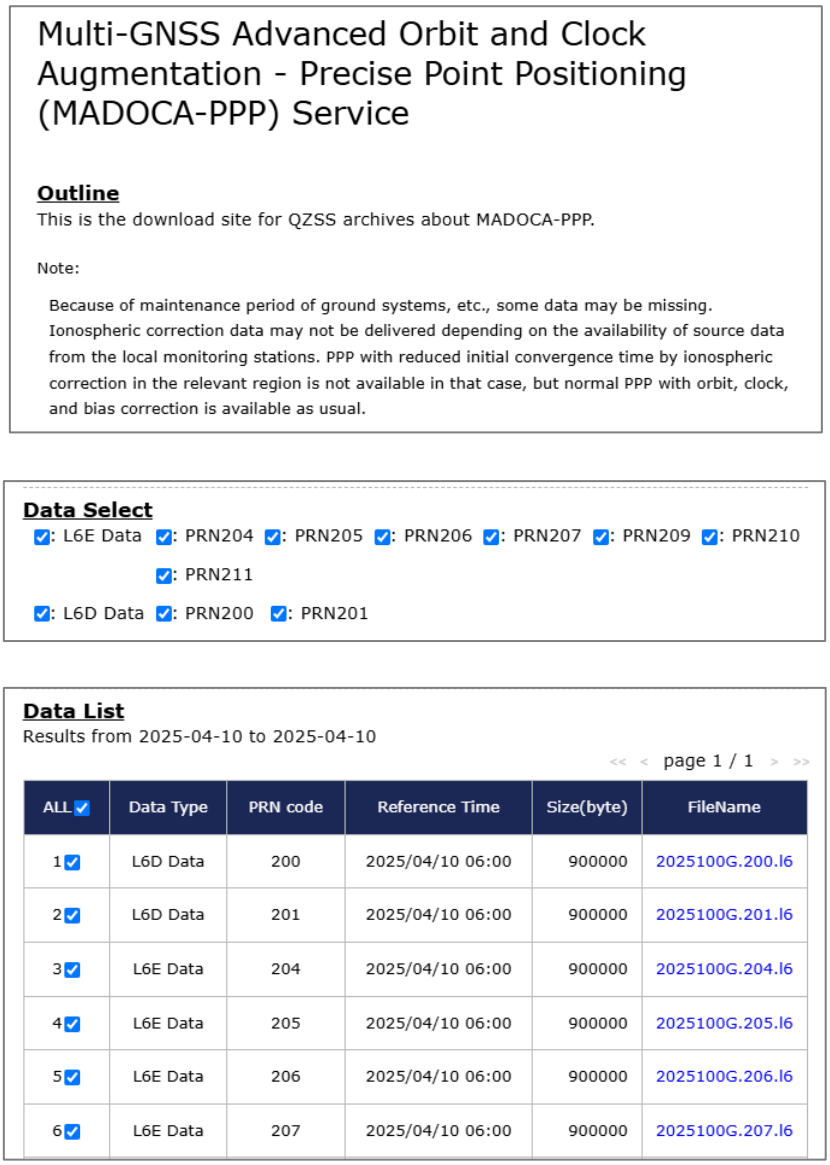
For Windows 64-bit environment, download RNXCMP_4.1.0_Windows_mingw_64bit.

(2) Getting QZSS L6 message files

L6E and L6D archive data file can be downloaded from the following QZSS URL.

https://sys.qzss.go.jp/dod/en/archives.html

Hourly files can be obtained for each PRN number. To perform PPP/PPP-AR in the Asia Oceania region, download the PRN204 and PRN206 files. For further ionospheric correction, please download the PRN200 and PRN201 files.



Multi-GNSS Advanced Orbit and Clock Augmentation - Precise Point Positioning (MADOCA-PPP) Service

Outline

This is the download site for QZSS archives about MADOCA-PPP.

Note:

Because of maintenance period of ground systems, etc., some data may be missing. Ionospheric correction data may not be delivered depending on the availability of source data from the local monitoring stations. PPP with reduced initial convergence time by ionospheric correction in the relevant region is not available in that case, but normal PPP with orbit, clock, and bias correction is available as usual.

Data Select

☑: L6E Data ☑: PRN204 ☑: PRN205 ☑: PRN206 ☑: PRN207 ☑: PRN209 ☑: PRN210 ☑: PRN211

☑: L6D Data ☑: PRN200 ☑: PRN201

Data List

Results from 2025-04-10 to 2025-04-10

<< < page 1 / 1 > >>

ALL ☑	Data Type	PRN code	Reference Time	Size(byte)	FileName
1 ☑	L6D Data	200	2025/04/10 06:00	900000	2025100G.200.l6
2 ☑	L6D Data	201	2025/04/10 06:00	900000	2025100G.201.l6
3 ☑	L6E Data	204	2025/04/10 06:00	900000	2025100G.204.l6
4 ☑	L6E Data	205	2025/04/10 06:00	900000	2025100G.205.l6
5 ☑	L6E Data	206	2025/04/10 06:00	900000	2025100G.206.l6
6 ☑	L6E Data	207	2025/04/10 06:00	900000	2025100G.207.l6

(3) Set up execution commands

Store the downloaded RINEX and L6 files in the following folders respectively.

`<installed_dir>¥sample_data¥data¥rinex¥obs¥<4digit year>¥<3digit day of year>`

`<installed_dir>¥sample_data¥data¥rinex¥BRDC¥<4digit year>¥<3digit day of year>`

`<installed_dir>¥sample_data¥data¥l6¥<4digit year>¥<3digit day of year>`

Open the batch file in the following folder with a text editor.

<*installed_dir*>¥sample_data¥exec_rnx2rtkp.bat

Rewrite the followings to specify the downloaded RINEX OBS, RINEX NAV, L6E message and L6D message.

```
OBS=data/rinex/obs/%Y/%n/MIZU00JPN_R_%Y%n0000_01D_30S_MO.rnx
NAV=data/rinex/BRDC/%Y/%n/BRDM00DLR_S_%Y%n0000_01D_MN.rnx
L6E1=data/l6/%Y/%n/%Y%n%HU.204.l6
L6E2=data/l6/%Y/%n/%Y%n%HU.206.l6
L6D1=data/l6/%Y/%n/%Y%n%HU.200.l6
L6D2=data/l6/%Y/%n/%Y%n%HU.201.l6
```

It is recommended that file paths be specified using keywords that are replaced to date/time. See 3.1.2.

Rewrite the following start date/time and end date/time according to the date and time of the analysis

```
TS_DATE=2026/03/28
TS_TIME=00:00:00
TE_DATE=2026/03/28
TE_TIME=23:59:30
```

Double-click the batch file to run it. And you can get output files in the following directory.

<*installed_dir*>¥sample_data¥data¥result_exec_rnx2rtkp

Note. In the batch file, **%** is replaced with **%%** to escape the special character **%** in Windows.

3 Usage of Utilities

3.1 RNX2RTKP

3.1.1 Overview

RNX2RTKP is a utility for post-process positioning. It implements a processing mode called PPP (Precise Point Positioning) to make use of MADOCALIB.

3.1.2 Inputs and Outputs

The input and output files for the RNX2RTKP are as follows.

Inputs

No.	Item	Description
1	RINEX OBS files	RINEX v3.x Observation files (*.rnx)(*1).
2	RINEX NAV files	RINEX v3.x Navigation files (*.rnx) (*1).
3	QZSS L6E files	L6E archive data file which can be downloaded from the following QZSS website (*.*PRN*.l6) (*1). PRN=204~211 https://sys.qzss.go.jp/dod/en/archives.html
4	QZSS L6D files	L6D archive data file which can be downloaded from the following QZSS website (*.*PRN*.l6) (*1). PRN=200,201 https://sys.qzss.go.jp/dod/en/archives.html
5	ANTEX file	The antenna phase information file (e.g. igs20.atx, igs14.atx) which can be obtained from the following IGS website.(*For monitoring stations that do not use the same combination of antenna and radome used for IGS stations, a dedicated antenna phase information file needs to be used instead of the above file.) https://cddis.nasa.gov/Data_and_Derived_Products/GNSS/GNSS_product_holdings.html This file is only used for receive antenna correction.
6	Configuration file	Configuration file of RNX2RTKP. See 3.1.3.

Outputs

No.	Item	Description
1	Position Solution file	File containing the receiver's position as calculated by the PPP process (*1). The file is a simple text file, and its format is defined in the RTKLIB manual “B.1 Positioning Solution File” at the following URL. https://github.com/tomojitakasu/RTKLIB/blob/master/doc/manual_2.4.2.pdf

(*1)
These file path can be specified with the keywords replaced as below

```
%Y -> yyyy year (4 digits) (2000-2099)
%y -> yy year (2 digits) (00-99)
%m -> mm month (01-12)
%d -> dd day of month (01-31)
%h -> hh hours (00-23)
%H -> a hour code (a-x)
%M -> mm minutes (00-59)
%n -> ddd day of year (001-366)
%W -> wwww gps week (0001-9999)
%D -> d day of gps week (0-6)
%HU-> h hour code (A=0,B=1,C=2,...,X=23)
```

3.1.3 **Configuration file**

The keywords which can be included in the configuration file for RNX2RTKP are shown in the following tables.

Item	Description	Configuration File	Notes
Positioning Mode	Set the positioning mode. MADOCALIB supports following options. - `ppp-kine` : PPP with kinematic mode - `ppp-static` : PPP with static mode	`pos1-posmode`	
Number of Frequencies	Set the used number of frequencies - `l1` : Single frequency. N/A for `ppp-*` modes. - `l1+2` : Dual-frequency - `l1+2+3` : Triple-frequency - `l1+2+3+4` : Quad-frequency	`pos1-frequency`	

Item	Description	Configuration File	Notes
Ionosphere Correction	Set the ionospheric correction option. - `off` : Not apply ionospheric correction - `brdc` : Apply broadcast ionospheric correction. - `dual-freq` : Ionosphere-free linear combination with dual-frequency. This should be selected if PPP is performed. - `est-stec` : Estimate ionospheric delay for each satellite. This should be selected if PPP-AR is performed.	`pos1-ionoopt`	
Navigation System	Set the used number of navigation satellite systems. - `1` : GPS - `4` : GLONASS - `8` : Galileo - `16` : QZSS - `32` : BeiDou If you want to use PPP with GPS+GAL+QZS, you should set the sum of these numbers to "`25`" because GPS is "`1`", GAL is "`8`", and QZS is "`16`".	`pos1-navsys`	
Ionospheric Correction	Set apply for ionospheric correction by L6D. - `off` : Not apply ionospheric correction - `on` : apply ionospheric correction	`pos2-ionocorr`	L6D file must be specified as input file when applying the ionosphere correction.
Integer Ambiguity Resolution	Set the strategy of integer ambiguity resolution. - `off` : No ambiguity resolution - `continuous` : Continuously integer ambiguities are estimated and resolved - `fix-and-hold` : Continuously integer ambiguities are estimated and resolved. If the validation OK, the ambiguities are tightly constrained to the resolved values.	`pos2-armode`	
Validation Ratio to Fix Ambiguity	Set the integer ambiguity validation threshold for "ratio-test", which uses the ratio of squared residuals of the best integer vector to the second-best vector.	`pos2-arthres`	
Std-Dev for Ambiguity Search	Set the maximum standard deviation threshold of the 3D position to start narrow-lane integer ambiguity search (m).	`pos2-arthres1`	

Item	Description	Configuration File	Notes
Signal Option for GPS	Set the priority of used signals for GPS. - L1/L2 - L1/L5 - L1/L2/L5	pos2-siggps	The actual number of frequencies to be used shall be set by "pos1-frequency". If L1/L2/L5 are set and "pos1-frequency" is set to "l1+2", dual frequencies of L1/L2 are used. (*1)
Signal Option for QZSS	Set the priority of used signals for QZSS. - L1/L5 - L1/L2 - L1/L5/L2	pos2-sigqzs	Same as above.
Signal Option for Galileo	Set the priority of used signals for Galileo. - E1/E5a - E1/E5b - E1/E5a/E5b/E6 - E1/E5a/E6/E5b	pos2-siggal	Same as above.
Signal Option for BeiDou-2	Set the priority of used signals for BeiDou-2. - B1I/B3I - B1I/B2I - B1I/B3I/B2I	pos2-sigbds2	Same as above.
Signal Option for BeiDou-3	Set the priority of used signals for BeiDou-3. - B1I/B3I - B1I/B2a - B1I/B3I/B2a	pos2-sigbds3	Same as above.
BDS-2 Satellites in PRN 1-18	Specify the satellites in PRN 1-18 to be treated as BDS-2 after the BeiDou PRN reassignment of April 2026. Other satellites are treated as BDS-3. Satellite IDs (C01-C18) may be specified as a comma-separated list or ranges (e.g., C09-C10). For L6 messages recorded before the reassignment, specify C01-C18.	pos2-bds2prn	

Item	Description	Configuration File	Notes
BDS-3 IGSO Satellites	Specify the BDS-3 IGSO satellites. They are not selected as the reference satellite of the integer ambiguity resolution. Satellite IDs may be specified as a comma-separated list or ranges (e.g., C06-C08). If empty, C06-C08 are used, which are the BDS-3 IGSO satellites after the BeiDou PRN reassignment of April 2026. For L6 messages recorded before the reassignment, specify C38-C40.	`pos2-bds3igso`	

(*1)

The signal option settings mean from left to right: 1st frequency, 2nd frequency, 3rd frequency, 4th frequency. Observation data for the 1st and 2nd frequencies must be required to perform PPP, while observation data for the 3rd and 4th frequencies are optional. Therefore, if the observed data for the frequencies set in 1st freq. and 2nd freq. are not present in one of them, PPP will not be performed properly.

For example, if the following options are set for a receiver outputting observation data of the L1 and L2 frequencies, PPP will be performed in dual-frequency mode internally.

```
pos1-frequency     =l1+2+3
pos2-siggps          =L1/L2/L5
```

However, with the same settings, PPP cannot be performed for observation data of L1 and L5 frequencies. In this case, it should be set up as follows.

```
pos2-siggps           =L1/L5
```

The following sample files are included on the directory as below.

<*installed_dir*>¥app¥consapp¥rnx2rtkp¥gcc_mingw

`sample.conf`

Configuration for PPP (no integer ambiguity resolution) whose performance is specified in PS-QZSS. This configuration performs PPP by ionosphere-free linear combination with dual-frequency with GPS, GLONASS, QZSS and Galileo by setting as below.

```
pos1-navsys        =29
pos1-frequency     =l1+2
pos1-ionoopt       =dual-freq
pos2-armode        =off
```

`sample_pppar.conf`

Configuration for PPP-AR (with integer ambiguity resolution). This configuration performs PPP-AR with quad-frequency with GPS, GLONASS, QZSS, Galileo and BeiDou by setting as below.

```
pos1-navsys        =61
pos1-frequency     =l1+2+3+4
pos2-armode        =continuous
```

`sample_pppar_iono.conf`

Configuration for PPP-AR with ionospheric correction on L6D message. This configuration performs PPP-AR with ionospheric correction by setting as below.

```
pos2-ionocorr      =on
```

These sample configuration files are tuned specifically for PPP processing for observation data at 30-second intervals at MIRAI or IGS stations, using the MADOCA-PPP L6 message archive data as augmentation data. Therefore, it is recommended that the parameters especially `pos2-arthres` and `pos2-arthres1` be tuned accordingly for each case where observation data conditions such as data intervals are different.

3.1.4 **CUI Command Reference**

```
synopsis:
  rnx2rtkp.exe [option]... obsfile navfile l6efile l6dfile [...]

options:
 -k file   input options from configuration file [off]
 -o file   set output file [stdout]
 -ts ds ts start day/time (ds=y/m/d ts=h:m:s) [obs start time]
 -te de te end day/time   (de=y/m/d te=h:m:s) [obs end time]
 -ti tint  time interval (sec) [all]
 -x level  debug trace level (0:off) [0]
 -ant file rcvantfile [specified by conf file]

example:
  rnx2rtkp.exe -ts 2025/04/01 00:00:00 -te 2025/04/01 23:59:30 -ti 30
               -k sample_pppar_iono.conf -o test.%Y%m%d%h%M%S.pos
                MIZU00JPN_R_%Y%n0000_01D_30S_MO.rnx
                BRDM00DLR_S_%Y%n0000_01D_MN.rnx
                %Y%n%HU.209.l6 %Y%n%HU.200.l6 %Y%n%HU.201.l6
```

3.2 **CSSR2SSR**

3.2.1 **Overview**

CSSR2SSR is a utility for post-processing conversion of L6E message file containing MADOCA-PPP compact SSR data to RTCM3 SSR message file.
Convert cssr message file to ssr message file. Supported Compact SSR messages (cssr) and SSR messages (ssr) are as bellow.

```
[Vendor ID]
  "010b" : MADOCA-PPP

[Message Type, Sub Type]
  MT4073,1 : mask
  MT4073,2 : orbit
  MT4073,3 : clock
  MT4073,4 : code bias
  MT4073,5 : phase bias
  MT4073,7 : ura

Supported SSR messages (ssr) are as bellow.

[Message Type]
  SSR Message          : GPS       GLONASS    Galileo       QZSS        BeiDou
  --------------------:-----------------------------------------------------
  SSR Orbit Correction: 1057      1063       1240(*1)     1246(*1)    1258(*1)
      Clock Correction: 1058      1064       1241(*1)     1247(*1)    1259(*1)
      URA             : 1061      1067       1244(*1)     1250(*1)    1262(*1)
      Code Bias       : 1059(*2)  1065(*2)   1242(*1*2)   1248(*1)    1260(*1)
      Phase Bias      : 1265(*2)  1266(*2)   1267(*1*2)   1268(*1*2)  1270(*1*2)

[format]
  nothing * :RTCM 10403.3
  (*1)      :Proposal of new RTCM SSR Messages
             SSR Stage 1:Galileo,QZSS,SBAS,BDS 2014-04-17 v05
  (*2)      :Proposal of new RTCM SSR Messages
             SSR Stage 2:Satellite Phase Biases 2014-04-17 v05
```

The signals stored in MADOCA-PPP that are not defined in RTCM SSR utilize the following signal and tracking mode identifiers.

```
GNSS    :Signal and Tracking Mode Identifier
  --------:----------------------------------
  GPS     :16- L5 I+Q
          :17- L1 L1C(D+P)
  Galileo : 3- E1 B+C
          : 7- E5a I+Q
          :10- E5b I+Q
  BeiDou  :13- B2a Q
          :14- B2a I+Q
```

3.2.2 **Inputs and Outputs**

The input and output files for the CSSR2SSR are as follows.

Inputs

No.	Item	Notes
1	QZSS L6E file	L6E message file containing MADOCA-PPP compact SSR (CSSR) data.

Outputs

No.	Item	Notes
1	RTCM3 SSR message file	RTCM3 SSR message file.

3.2.3 **Configuration file**

There is no configuration file for the CSSR2SSR.

3.2.4 **CUI Command Reference**

```
synopsis:
  cssr2ssr.exe [options] file

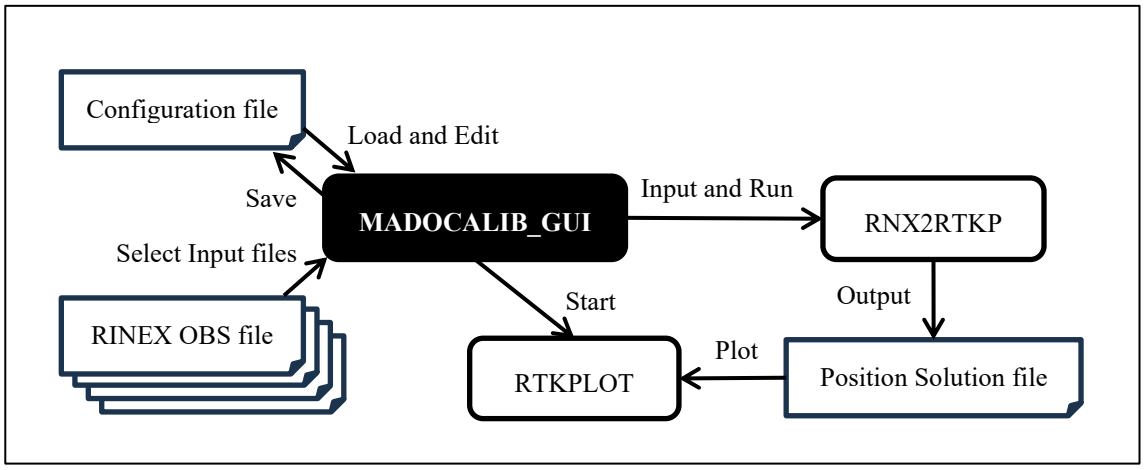
options:
  -td y/m/d      date for message time (y=year,m=month,d=day)
  -o file        output ssr file
  -d             dump debug log on decoding cssr [default : no dump]
  -bds2prn list  BDS-2 satellites in PRN 1-18 (GNSS ID 3), e.g. C09,C10
                 (other satellites in GNSS ID 3 are BDS-3) [default : no BDS-2]
                 For L6 messages recorded before the reassignment,
                 specify C01-C18.
  file           input l6E message file

example:
  cssr2ssr.exe -td 2025/04/01 -bds2prn C01-C18 -o 202509100.204.rtcm3
               202509100.204.l6
```

3.3 **MADOCALIB_GUI**

3.3.1 **Overview**

MADOCALIB_GUI is a Windows GUI application that enables users to configure and execute the console-based RNX2RTKP, and to start RTKPLOT to visualize the processing results.
For an overview of the MADOCALIB_GUI workflow, follow the conceptual diagram below.



3.3.2 **Inputs and Outputs**

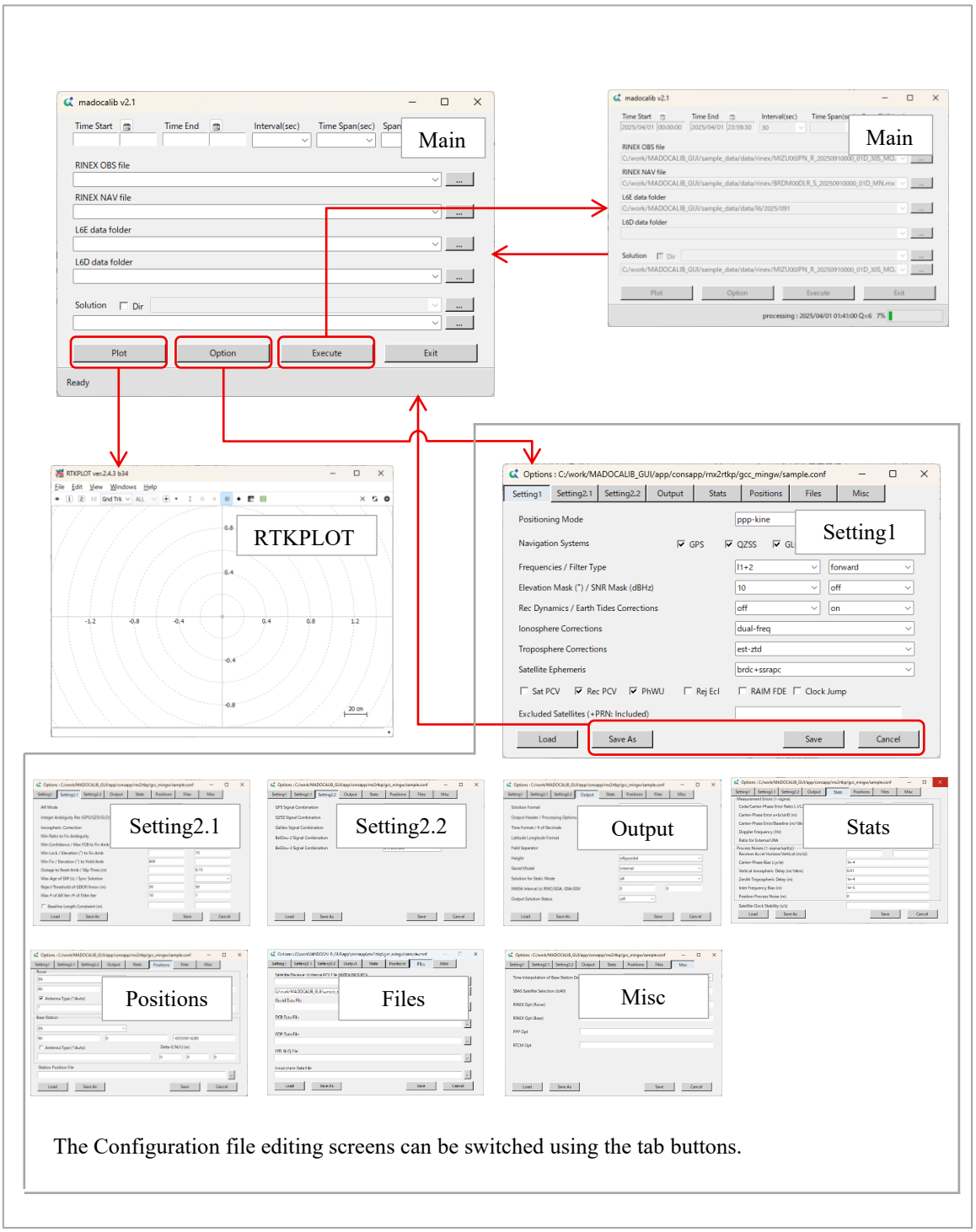
The inputs and outputs follow the definitions provided in the RNX2RTKP application.
For details, see 3.1.2.

3.3.3 **Configuration file**

The configuration file follows the definitions provided in the RNX2RTKP application.
For details, see 3.1.3.

3.3.4 **Screen Flow**

The screen flow of MADOCALIB_GUI is as follows:



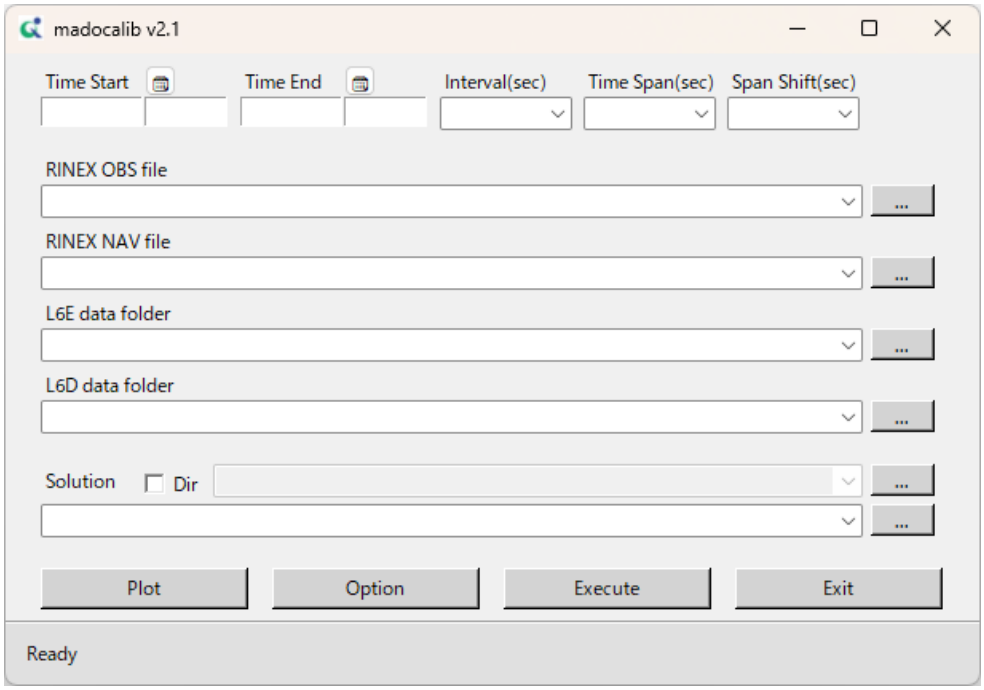
The Configuration file editing screens can be switched using the tab buttons.

3.3.5 **Screen Descriptions**

(1) Main

This is the first screen displayed when start MADOCALIB_GUI. From this screen, It is possible to perform the following actions:

- Select various input files for RNX2RTKP
- Specify the output directory
- Configure the start/end time, interval, and positioning span
- Start RTKPLOT
- Open the Configuration file editor
- Execute RNX2RTKP
- Exit the application



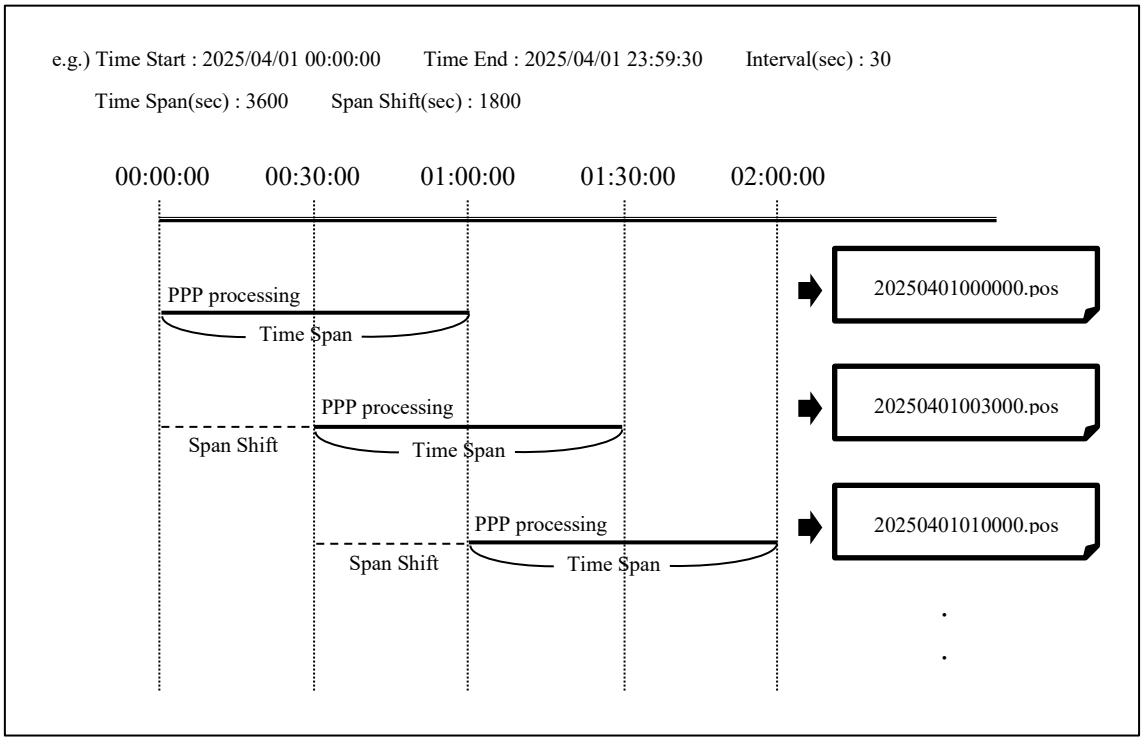
Screen Area	Descriptions
Time Start	Required field. The start date and time of the positioning process. When a RINEX OBS file is selected, the value is filled automatically; manual input is also possible. The date/time can be selected from the GUI using the calendar button. It should be earlier than Time End.

Screen Area	Descriptions
Time End	Required field. The end date and time of the positioning process. When a RINEX OBS file is selected, the value is filled automatically; manual input is also possible. The date/time can be selected from the GUI using the calendar button. It should be later than Time Start.
Interval(sec)	Required field. The execution interval of the positioning process in seconds. When a RINEX OBS file is selected, the value is filled automatically; manual input is also possible.
Time Span(sec)	Optional field. The duration of a single positioning process in seconds. When this option is set, positioning is performed repeatedly from Time Start to Time End at the specified span interval. If repeated processing is executed, multiple .pos files are generated. The generated .pos files are stored in a folder named <*result_exec*>, automatically created in the same directory as the path specified in Solution. Each .pos filename corresponds to the start date and time of the individual processing span (YYYYMMDDhhmmss.pos). Details are shown in "Operational Image of Time Span and Span Shift".
Span Shift(sec)	Optional field. The interval between spans in seconds, used together with Time Span. If left blank, the value is treated as identical to Time Span. Details are shown in "Operational Image of Time Span and Span Shift".
RINEX OBS file	Required field. The file can be selected using the dialog button or by dragging and dropping it into the text area.
RINEX NAV file	Required field. The file can be selected using the dialog button or by dragging and dropping it into the text area.

Screen Area	Descriptions
L6E data folder	Required field. The folder containing L6E files can be selected using the dialog button or by dragging and dropping it into the text area.
L6D data folder	Optional field. The folder containing L6D files can be selected using the dialog button or by dragging and dropping it into the text area.
Solution	Required field. When a RINEX OBS file is selected, the value is filled automatically; manual input is also possible. An output path or folder can be selected using the dialog button or by dragging and dropping an item into the text area. When Dir is unchecked, a full output file path should be specified. When Dir is checked, an output directory and filename should be specified.
Plot	Starts RTKPLOT. If a pos file is specified in Solution, it is automatically loaded.
Option	Opens the screen for editing the configuration file. When the cursor is placed over the Option button, a tooltip displays the path of the currently loaded configuration file.
Execute	Executes RNX2RTKP using the current settings.
Exit	Exits the application. The current settings are saved upon exit.

Screen Area	Descriptions
Status bar	Displays the status of the application. The progress of RNX2RTKP is shown in the lower-right area of the status bar. Messages in the lower-left area include: **Ready** The application is idle. **Done!** The positioning process has completed successfully. **Conf saved!** The configuration file has been saved. **madocalib: {any}** Standard error output from RNX2RTKP. **No .l6 files found in the selected folder. Please check and try again.** It is expected that the folder should contain L6E/L6D files. **Warning: path error load {path}** The configuration file could not be loaded. **Warning: No such conf file.** The specified configuration file does not exist. **No such {path}** RTKPLOT or RNX2RTKP executable not found. **No such OBS file** No OBS file has been specified. **No such NAV file** No NAV file has been specified. **No such L6E files** No L6E file has been specified. **Error Start/End Date** Time Start or Time End has not been specified. **Error Interval** Interval has not been specified.

Operational Image of Time Span and Span Shift

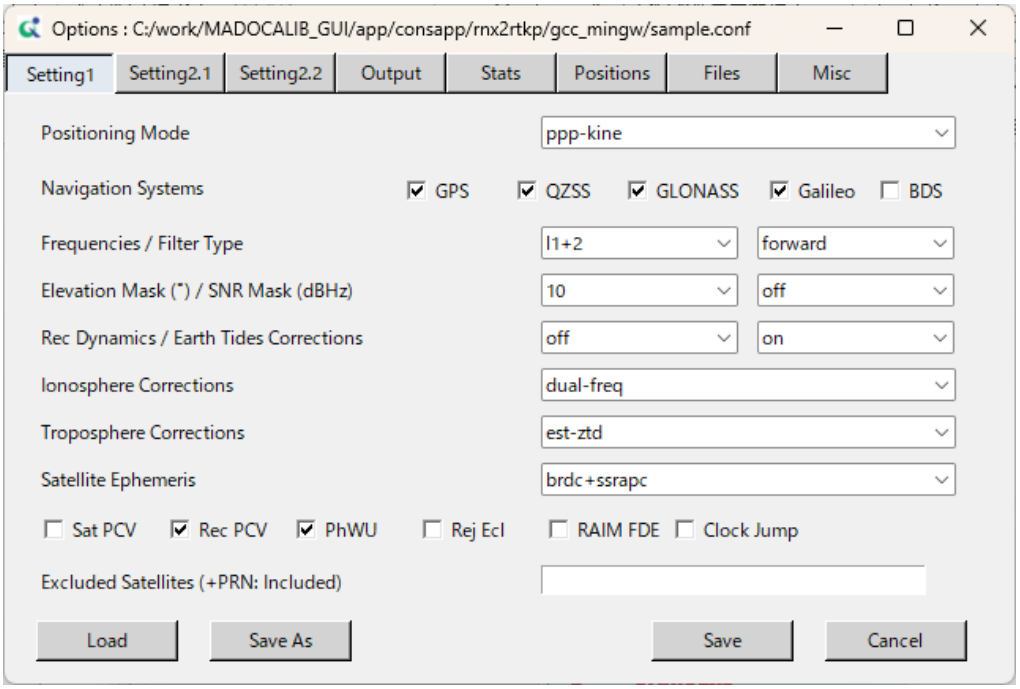


(2) Setting1

This screen allows you to edit the Configuration file.

It is possible to configure basic positioning parameters such as positioning mode and frequency settings.

Across all Configuration file editing screens, It is possible to load another Configuration file, overwrite the current file, or save it under a new name.



Screen Area	Descriptions
Screen Tab	A tab button used to switch between configuration screens. Each screen provides access to a different set of parameters.
Save	Overwrites the currently loaded configuration file with the latest settings.
Cancel	Closes the current screen.
Load	Loads another configuration file.
Save As	Saves the current settings under a new file name.
Configuration Edit	Refer to the following table for details on the parameters available in each configuration file.

Except for the functions within Configuration Edit, the Save and Load operations behave in the same way across all screens.

GUI Item	Configuration file	Notes
Positioning Mode	pos1-posmode	
Navigation Systems	pos1-navsys	
Frequencies	pos1-frequency	

GUI Item	Configuration file	Notes
Filter Type	pos1-soltype	
Elevation Mask (°)	pos1-elmask	
SNR Mask (dBHz)	pos1-snrmask_r	
Rec Dynamics	pos1-dynamics	
Earth Tides Corrections	pos1-tidecorr	
Ionosphere Corrections	pos1-ionoopt	
Troposphere Corrections	pos1-tropopt	
Satellite Ephemeris	pos1-sateph	
Sat PCV	pos1-posopt1	
Rec PCV	pos1-posopt2	
PhWU	pos1-posopt3	
Rej Ecl	pos1-posopt4	
RAIM FDE	pos1-posopt5	
Clock Jump	pos1-posopt6	
Excluded Satellites (+PRN: Included)	pos1-exclsats	

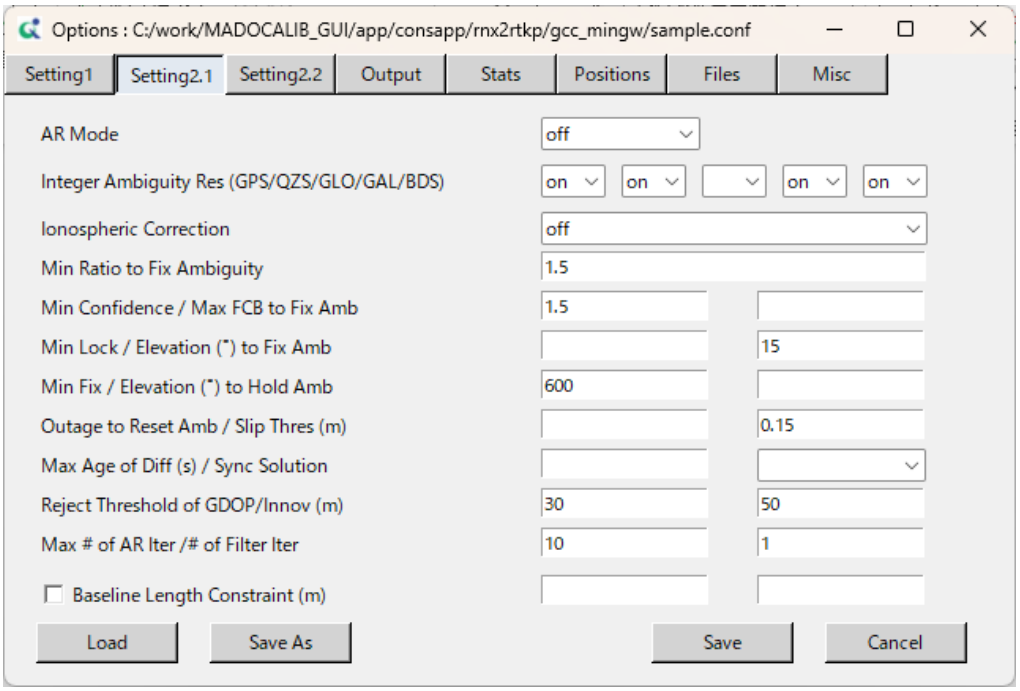
The configuration file follows the definitions provided in the RNX2RTKP application.

For details, see 3.1.3.

(3) Setting2.1

This screen allows you to edit the Configuration file.

It is possible to enable or disable Ambiguity Resolution (AR), select satellite systems, and configure AR-related thresholds.



GUI Item	Configuration file	Notes
AR Mode	pos2-armode	
Integer Ambiguity Res (GPS)	pos2-arsys	
Integer Ambiguity Res (QZS)	pos2-arsys	
Integer Ambiguity Res (GLO)	pos2-gloarmode	
Integer Ambiguity Res (GAL)	pos2-arsys	
Integer Ambiguity Res (BDS)	pos2-arsys	
Ionospheric Correction	pos2-ionocorr	
Min Ratio to Fix Ambiguity	pos2-arthres	
Min Confidence	pos2-arthres1	
Max FCB to Fix Amb	pos2-arthres2	Not used by RNX2RTKP.
Min Lock	pos2-arlockcnt	Not used by RNX2RTKP.
Elevation (°) to Fix Amb	pos2-arelmask	
Min Fix	pos2-arminfix	

GUI Item	Configuration file	Notes
Elevation (°) to Hold Amb	pos2-elmaskhold	
Outage to Reset Amb	pos2-aroutcnt	
Slip Thres (m)	pos2-slipthres	
Max Age of Diff (s)	pos2-maxage	Not used by RNX2RTKP.
Sync Solution	pos2-syncsol	Not used by RNX2RTKP.
Reject Threshold of GDOP	pos2-rejgdop	
Innov (m)	pos2-rejionno	
Max # of AR Iter	pos2-armaxiter	
# of Filter Iter	pos2-niter	
Baseline Length Constraint (m)	pos2-baselen	Not used by RNX2RTKP.
Baseline Length Constraint (m)	pos2-basesig	Not used by RNX2RTKP.

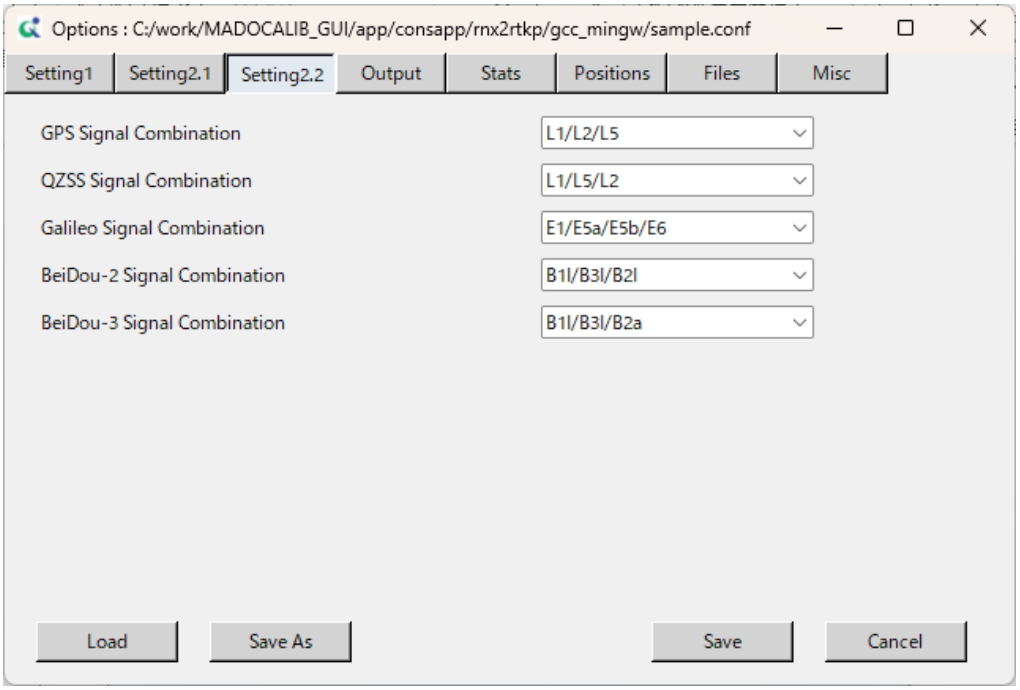
The configuration file follows the definitions provided in the RNX2RTKP application.

For details, see 3.1.3.

(4) Setting2.2

This screen allows you to edit the Configuration file.

It is possible to select the signals and frequencies to be used for each GNSS system.



GUI Item	Configuration file	Notes
GPS Signal Combination	pos2-siggps	
QZSS Signal Combination	pos2-sigqzs	
Galileo Signal Combination	pos2-siggal	
BeiDou-2 Signal Combination	pos2-sigbds2	
BeiDou-3 Signal Combination	pos2-sigbds3	
BeiDou-2 Satellites in PRN 1-18	pos2-bds2prn	
BeiDou-3 IGSO Satellites	pos2-bds3igso	

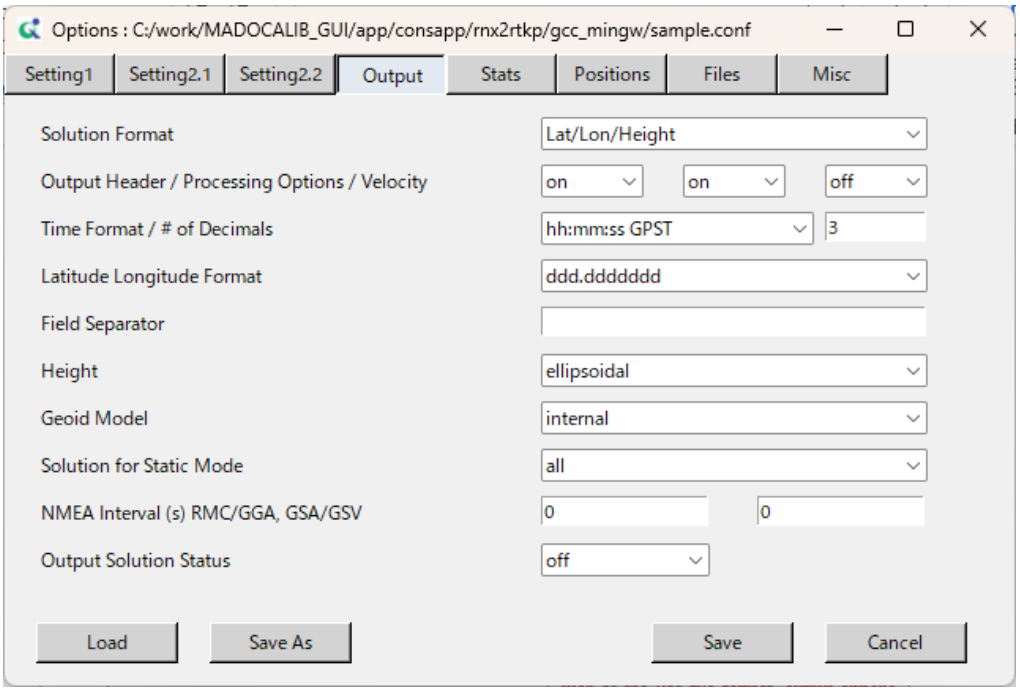
The configuration file follows the definitions provided in the RNX2RTKP application.

For details, see 3.1.3.

(5) Output

This screen allows you to edit the Configuration file.

It enables you to configure output settings such as the Pos file format, output options, and time systems/time formats.



GUI Item	Configuration file	Notes
Solution Format	out-solformat	
Output Header	out-outhead	
Processing Options	out-outopt	
Velocity	out-outvel	
Time Format	out-timesys	
# of Decimals	out-timendec	
Latitude Longitude Format	out-degform	
Field Separator	out-fieldsep	
Height	out-height	
Geoid Model	out-geoid	
Solution for Static Mode	out-solstatic	
NMEA Interval (s) RMC/GGA	out-nmeaintv1	
NMEA Interval (s) GSA/GSV	out-nmeaintv2	
Output Solution Status	out-outstat	

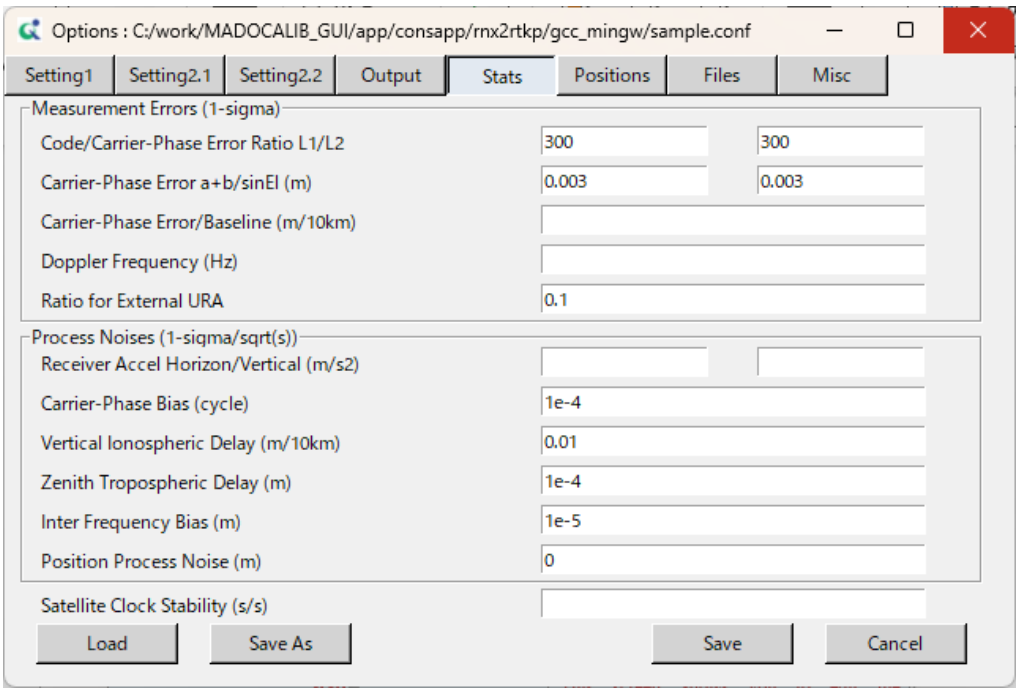
The configuration file follows the definitions provided in the RNX2RTKP application.

For details, see 3.1.3.

(6) Stats

This screen allows you to edit the Configuration file.

It is possible to configure parameters related to estimation, such as carrier-phase bias settings.



GUI Item	Configuration file	Notes
Code/Carrier-Phase Error Ratio L1/L2	stats-eratio1	
Code/Carrier-Phase Error Ratio L1/L2	stats-eratio2	
Carrier-Phase Error a+b/sinEl (m)	stats-errphase	
Carrier-Phase Error a+b/sinEl (m)	stats-errphaseel	
Carrier-Phase Error/Baseline (m/10km)	stats-errphasebl	Not used by RNX2RTKP.
Doppler Frequency (Hz)	stats-errdoppler	
Ratio for External URA	stats-uraratio	
Receiver Accel Horizon (m/s2)	stats-prnaccelh	
Receiver Accel Vertical (m/s2)	stats-prnaccelv	
Carrier-Phase Bias (cycle)	stats-prnbias	

GUI Item	Configuration file	Notes
Vertical Ionospheric Delay (m/10km)	stats-prniono	
Zenith Tropospheric Delay (m)	stats-prntrop	
Inter Frequency Bias (m)	stats-prnifb	
Position Process Noise (m)	stats-prnpos	
Satellite Clock Stability (s/s)	stats-clkstab	Not used by RNX2RTKP.

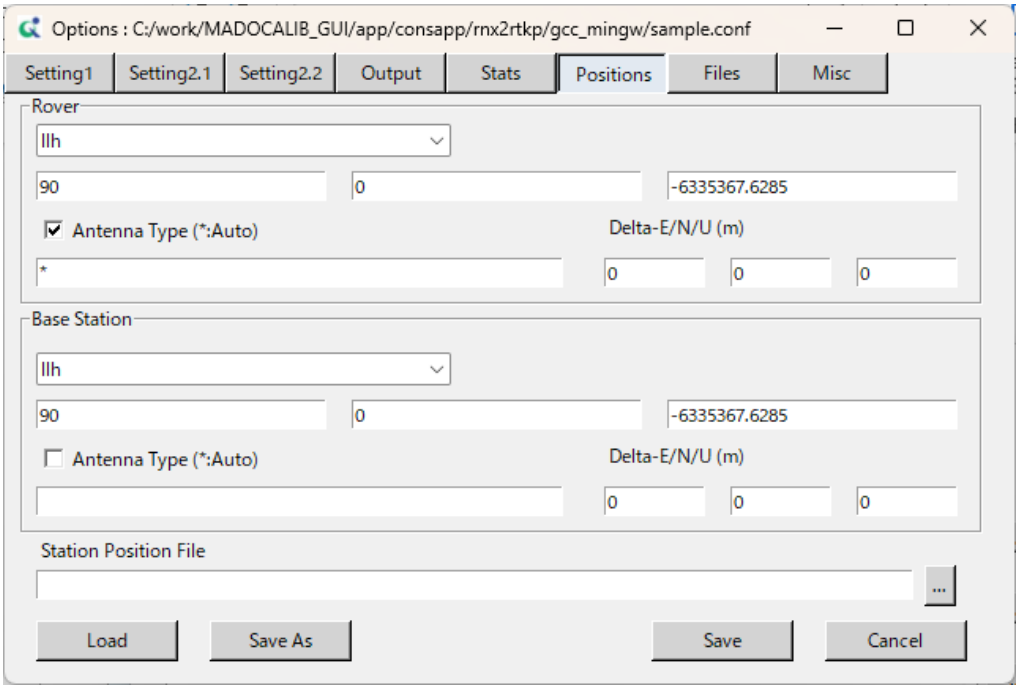
The configuration file follows the definitions provided in the RNX2RTKP application.

For details, see 3.1.3.

(7) Positions

This screen allows you to edit the Configuration file.

It is possible to configure the receiver antenna position input method, antenna type, and related settings.



GUI Item	Configuration file	Notes
Rover	ant1-postype	
Rover	ant1-pos1	
Rover	ant1-pos2	
Rover	ant1-pos3	
Antenna Type (*:Auto)	ant1-anttype	
Antenna Type (*:Auto)	ant1-anttype	
Delta-E/N/U (m)	ant1-antdele	
Delta-E/N/U (m)	ant1-antdeln	
Delta-E/N/U (m)	ant1-antdelu	
Base Station	ant2-postype	
Base Station	ant2-pos1	
Base Station	ant2-pos2	
Base Station	ant2-pos3	
Antenna Type (*:Auto)	ant2-anttype	
Antenna Type (*:Auto)	ant2-anttype	
Delta-E/N/U (m)	ant2-antdele	
Delta-E/N/U (m)	ant2-antdeln	
Delta-E/N/U (m)	ant2-antdelu	

GUI Item	Configuration file	Notes
Station Position File	file-staposfile	

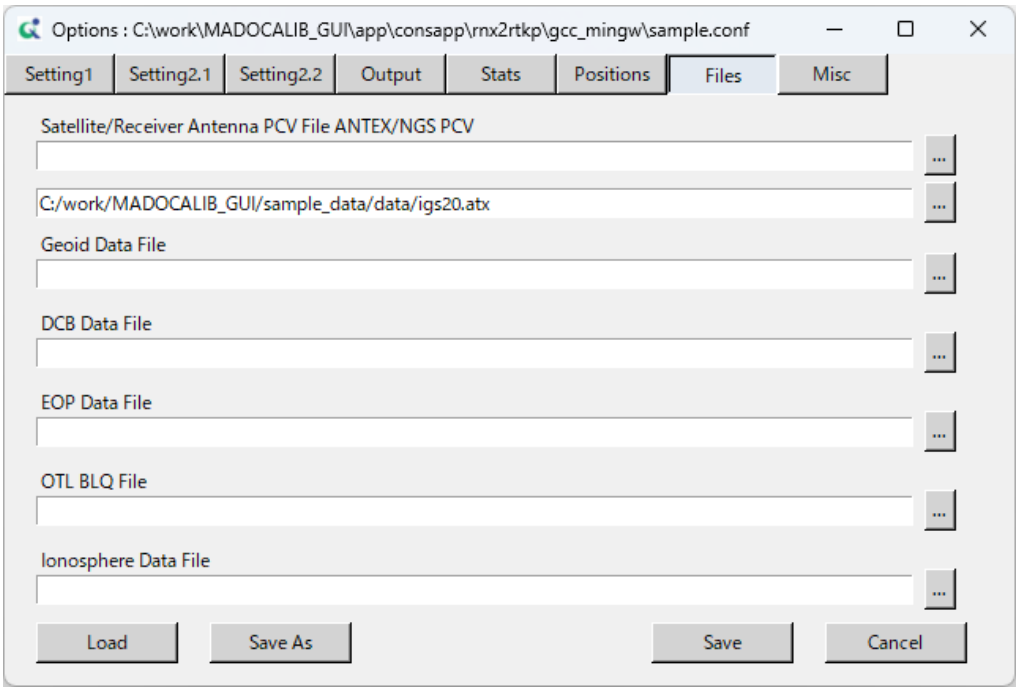
The configuration file follows the definitions provided in the RNX2RTKP application.

For details, see 3.1.3.

(8) Files

This screen allows you to edit the Configuration file.

It is possible to specify external files used for positioning, such as ANTEX files.



GUI Item	Configuration file	Notes
Satellite/Receiver Antenna PCV File ANTEX/NGS PCV	file-satantfile	
Satellite/Receiver Antenna PCV File ANTEX/NGS PCV	file-rcvantfile	
Geoid Data File	file-geoidfile	
DCB Data File	file-dcbfile	
EOP Data File	file-eopfile	
OTL BLQ File	file-blqfile	
Ionosphere Data File	file-ionofile	

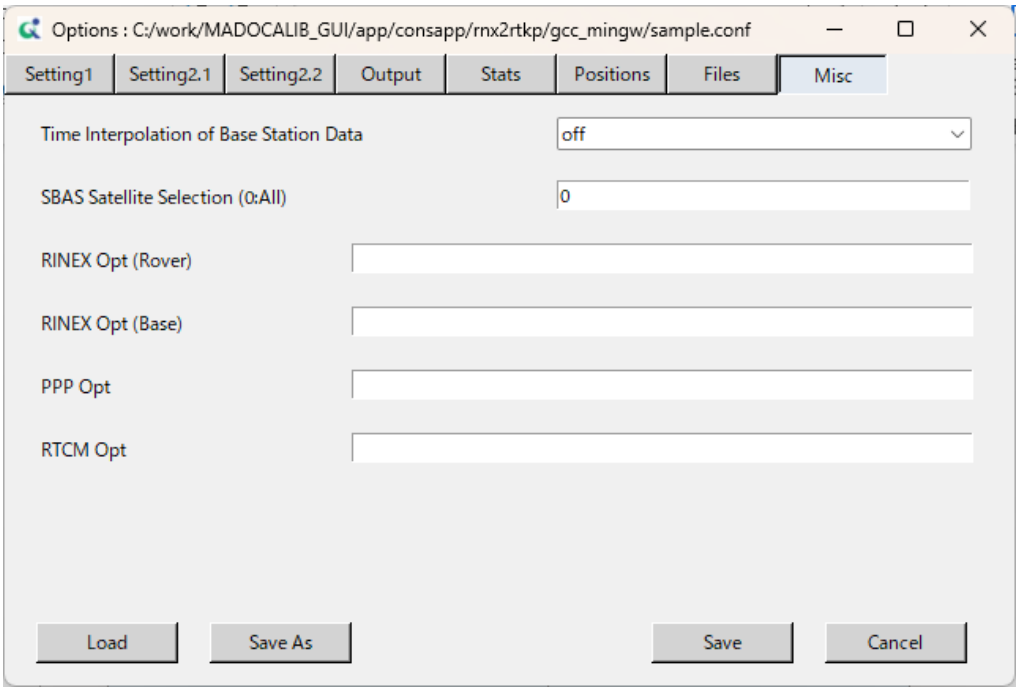
The configuration file follows the definitions provided in the RNX2RTKP application.

For details, see 3.1.3.

(9) Misc

This screen allows you to edit the Configuration file.

It is possible to configure auxiliary settings such as time interpolation of reference station data, in addition to the main parameters.



GUI Item	Configuration file	Notes
Time Interpolation of Base Station Data	misc-timeinterp	
SBAS Satellite Selection (0:All)	misc-sbasatsel	
RINEX Opt (Rover)	misc-rnxopt1	
RINEX Opt (Base)	misc-rnxopt2	
PPP Opt	misc-pppopt	
RTCM Opt	misc-rtcmopt	

The configuration file follows the definitions provided in the RNX2RTKP application.

For details, see 3.1.3.

3.3.6 **Quick Start with Sample Data**

Below is the Quick Start Guide for MADOCALIB_GUI.

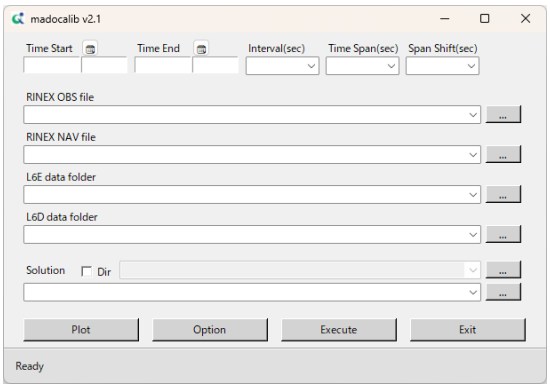
This guide demonstrates how to perform positioning using the bundled RNX2RTKP. Using the configuration files included in *<installed_dir>*¥sample_data, you will execute PPP (without integer ambiguity resolution).

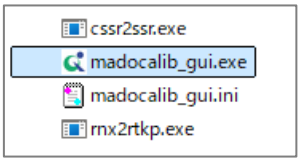
1) Running PPP Processing for a Full Day

Open the following folder in Windows Explorer:

<installed_dir>¥sample_data

Double-click madocalib_gui.exe to start the application.

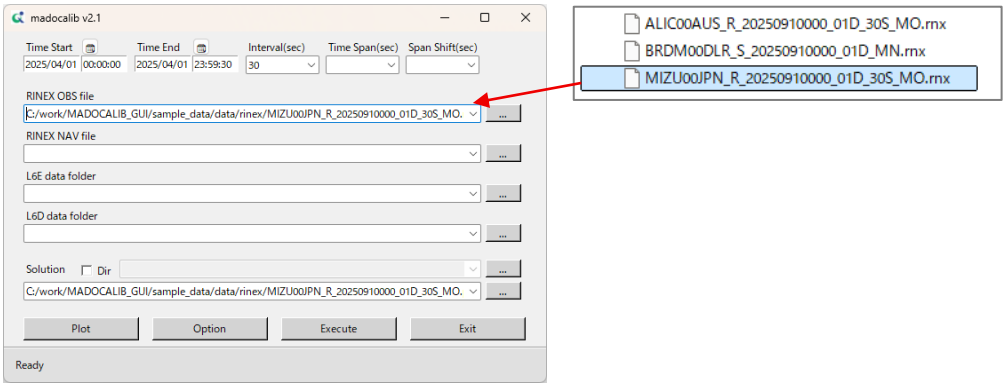




Drag and drop the RINEX OBS file located in

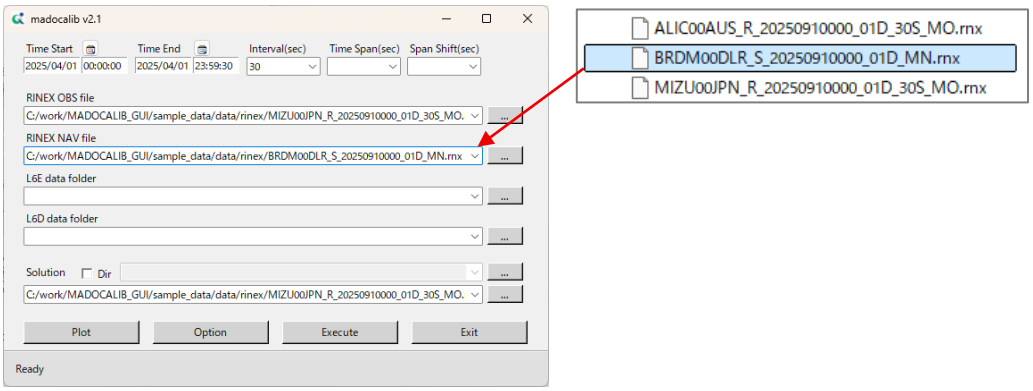
<installed_dir>¥sample_data¥data¥rinex¥obs¥2026¥087.

Time Start, Time End, Interval, and the output destination for the pos file will be filled in automatically.



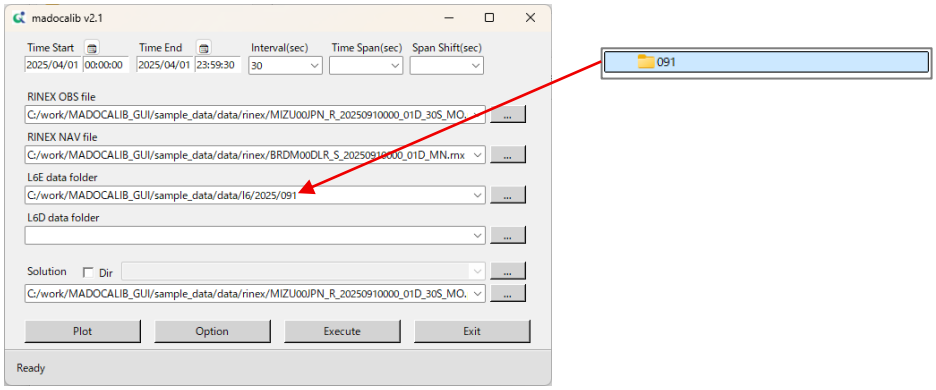
Drag and drop the RINEX NAV file located in

<installed_dir>¥sample_data¥data¥rinex¥BRDC¥2026¥087.



Drag and drop the folder <*installed_dir*>¥sample_data¥data¥l6¥2026¥087.

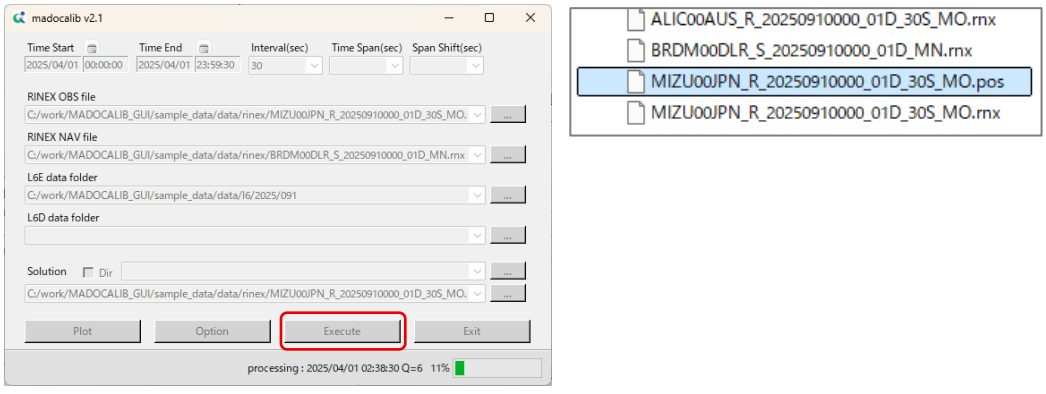
The 087 folder contains L6E data that corresponds to the Time Start and Time End range.



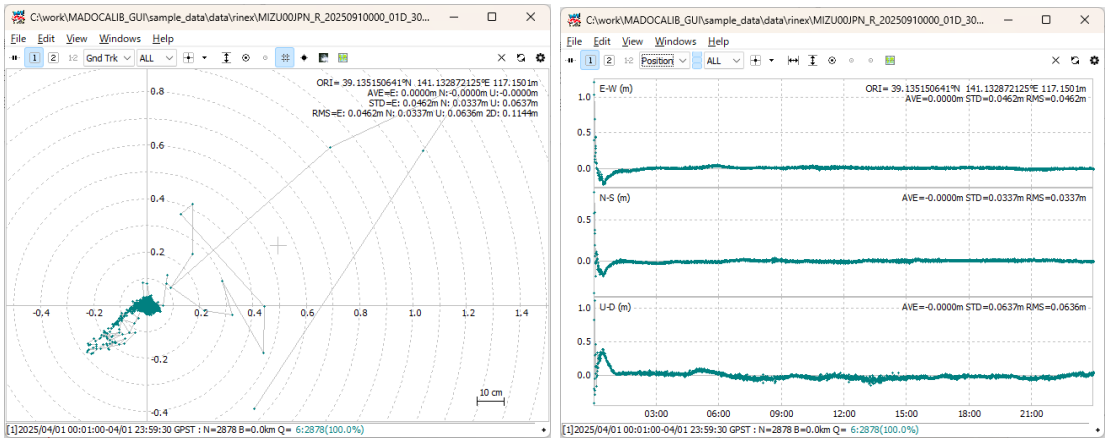
Click Execute.

The results for a full day will be generated.

`<installed_dir>¥sample_data¥data¥rinex¥obs¥2026¥087`



Click “PLOT” to start RTKPLOT. The generated .pos file will be plotted automatically.



2) Running PPP Processing Using Ionospheric Correction

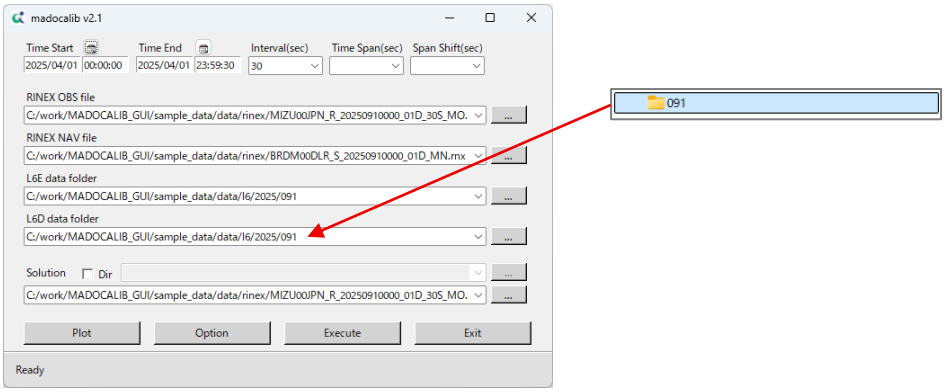
Drag and drop the RINEX OBS file MIZU00JPN_R_20250910000_01D_30S_MO.rnx located in <installed_dir>¥sample_data¥data¥rinex¥obs¥2025¥091.

Drag and drop the RINEX NAV file located in <installed_dir>¥sample_data¥data¥rinex¥BRDC¥2025¥091.

Drag and drop the folder <installed_dir>¥sample_data¥data¥l6_is-qzss-mdc-004¥2025¥091 into the L6E data folder box.

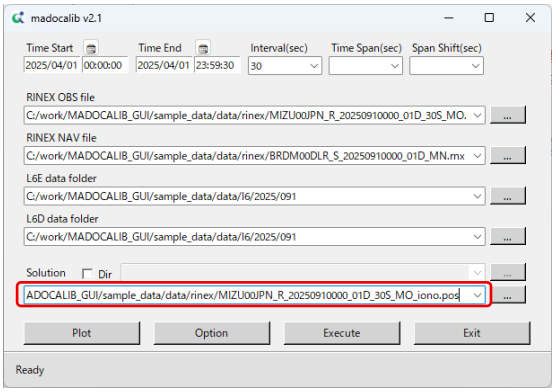
Drag and drop the folder <*installed_dir*>¥sample_data¥data¥l6_is-qzss-mdc-004¥2025¥091 into the L6D data folder box.

The 091 folder contains L6D data that corresponds to the Time Start and Time End range.

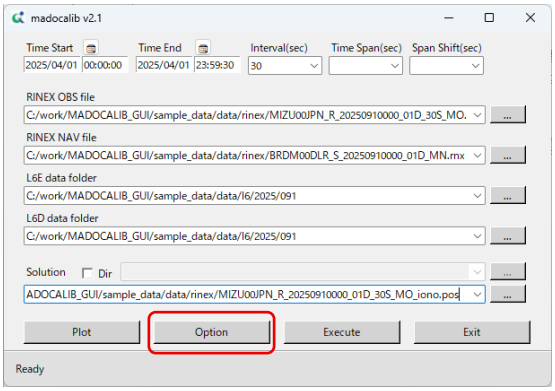


Enter any desired name for the Solution File.

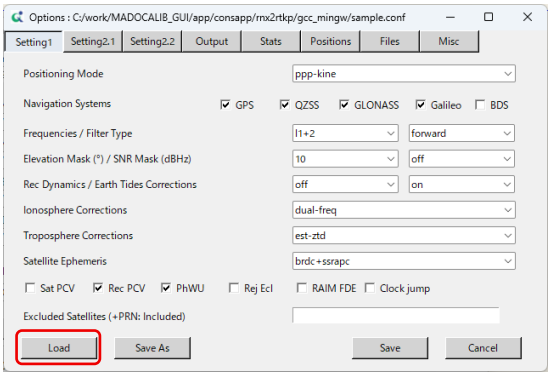
A different file name is used to avoid overwriting the previous file.



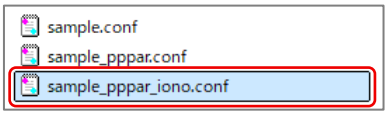
Click Option.



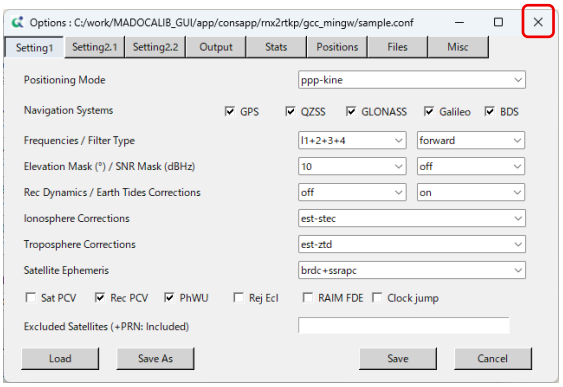
Click Load.



Select the <*installed_dir*>¥app¥consapp¥rnx2rtkp¥gcc_mingw¥sample_pppar_iono.conf in the file selection dialog and open it.



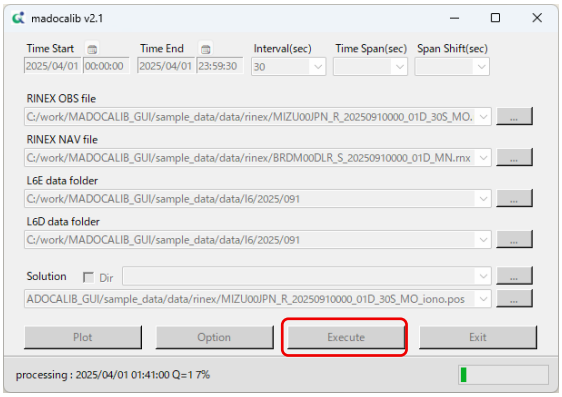
Click ×(Close).

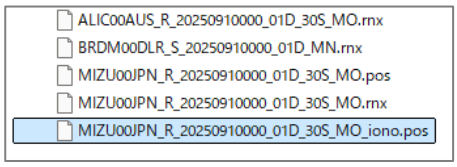


Click Execute.

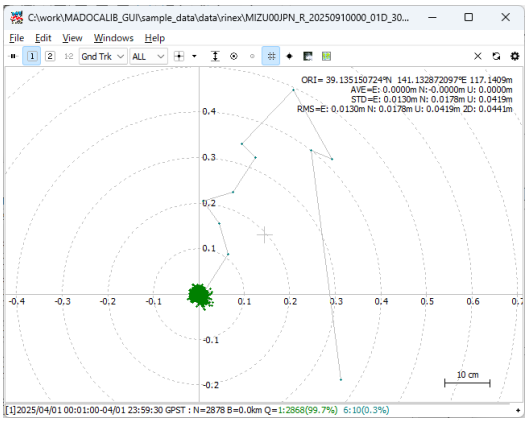
The results for a full day will be generated.

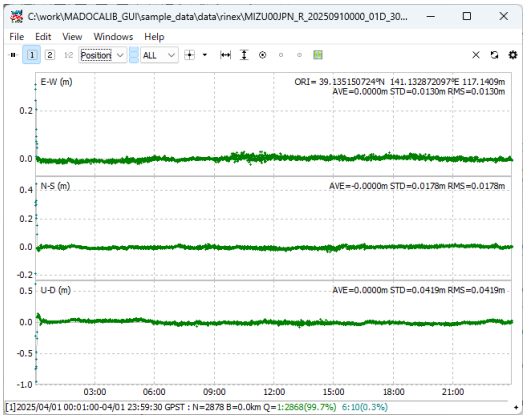
`<installed_dir>¥sample_data¥data¥rinex¥obs¥2026¥087`





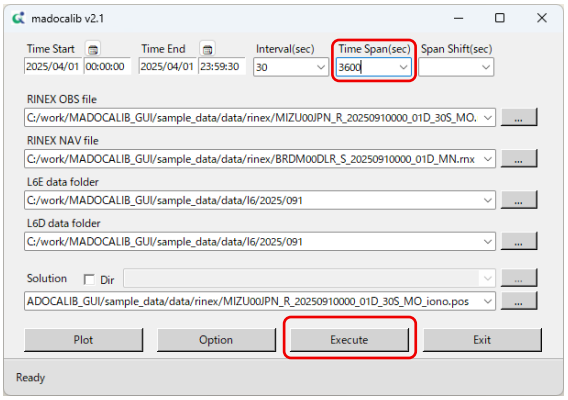
Click “PLOT” to start RTKPLOT. The generated .pos file will be plotted automatically.





3) Running PPP Processing at 1-Hour Intervals

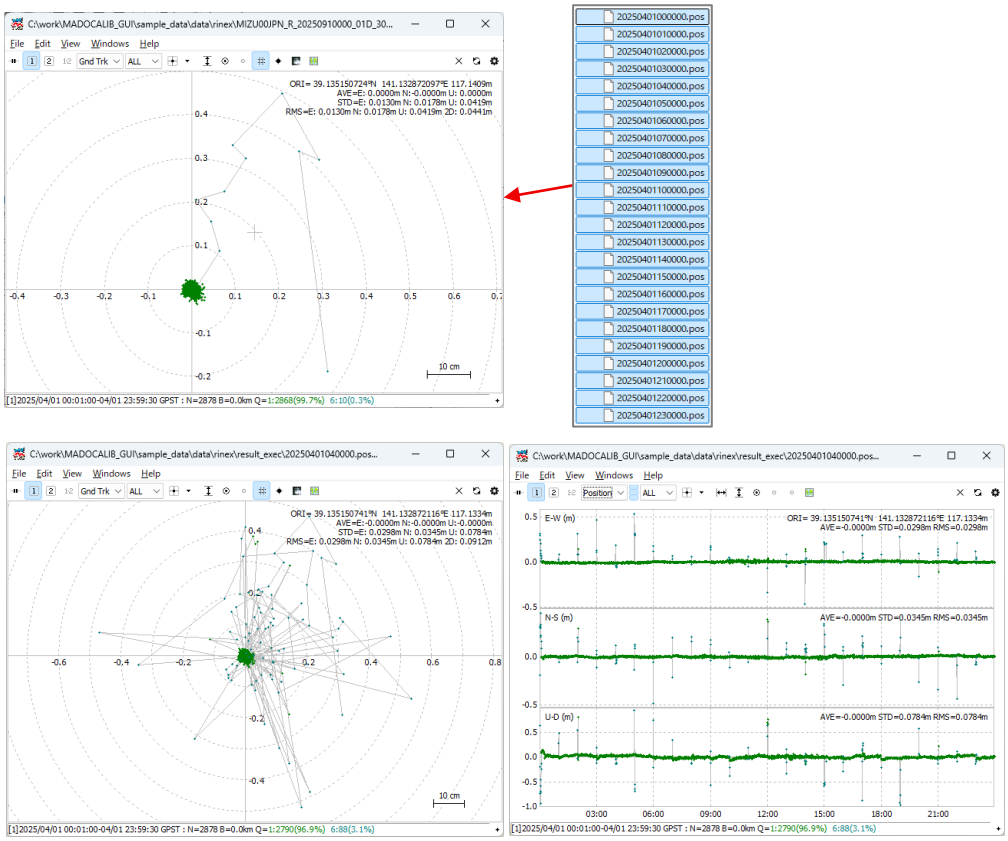
Enter 3600 (1 hour) in the Time Span field and click Execute.



A folder named <*result_exec*> will be created in the output directory.

One .pos file will be generated for each hour.

Drag and drop selected *.pos files into RTKPLOT to visualize the results.



Appendix. Specific Processing for MADOCA-PPP

This section outlines the specific processing of MADOCA-PPP within the MADOCALIB source files. In the source files located in the folder of `<installed_dir>¥src`, the following files are relevant to MADOCA-PPP processing.

(1) mdccssr.c

There are decode functions for MADOCA-PPP Compact SSR messages in QZSS L6E signal defined in chapter 4 of IS-QZSS-MDC. The main functions are as follows.

function	Description
`input_qzssl6e()`	Stack QZSS L6E message and synchronize frame with L6 preamble.
`decode_qzss_l6emsg()`	Decode QZSS L6E message and convert to RTCM SSR. Decode L6 message header, frame recognition, call decode functions of each subtypes.
`decode_mcssr_mask()`	Decode Sub Type 1 Mask.
`decode_mcssr_oc()`	Decode Sub Type 2 GNSS Orbit Corrections.
`decode_mcssr_cc()`	Decode Sub Type 3 GNSS Clock Corrections.
`decode_mcssr_cb()`	Decode Sub Type 4 GNSS Satellite Code Bias.
`decode_mcssr_pb()`	Decode Sub Type 5 GNSS Satellite Phase Bias.
`decode_mcssr_ura()`	Decode Sub Type 7 GNSS URA.
`mcssr_sel_biascode()`	Select MADOCA-PPP code bias, phase bias code from observation code. IS-QZSS-MDC section 5.5.3.1 Applicable Signals of Code/Phase Bias

(2) mdciono.c

There are decode functions for MADOCA-PPP ionospheric correction messages in QZSS L6D signal defined in section 6.3 of IS-QZSS-MDC. The main functions are as follows.

function	Description
`input_qzssl6d()`	Stack QZSS L6D message and synchronize frame with L6 preamble.
`decode_qzss_l6dmsg()`	Decode QZSS L6D message and store ionospheric corrections. Decode L6 message header, frame recognition, call decode functions of each message types.
`decode_miono_coverage()`	Decode Message Type 1 STEC coverage message.
`decode_miono_correction()`	Decode Message Type 2 STEC correction message.
`miono_get_corr()`	Get MADOCA-PPP ionospheric correction data using the user position.

(3) ppp.c

There are functions for PPP processing performed by EKF (extended Kalman filter) based on the observation equations defined in section 5.5 of IS-QZSS-MDC. The main functions are as follows.

function	Description
pppos()	PPP processing for observation data of an epoch. Execute time propagation (called "temporal update" in the source file) of EKF states, generate residuals of pseudorange and carrier phase, measurement update of EKF, and ambiguity resolution.
udstate_ppp()	Execute time propagation (temporal update) of EKF states such as receiver position, clocks, tropospheric parameters, ionospheric parameters, inter-frequency bias for 3rd/4th and phase bias.
corr_meas()	Pseudorange and carrier phase correction by SSR code/phase correction, antenna correction and phase wind-up correction.
ppp_res()	Generate residuals of pseudorange and carrier phase based on observation equations defined in section 5.5.5 of IS-QZSS-MDC.

(4) ppp_ar.c

There are functions for integer ambiguity resolution (AR). The MADOCALIB employs cascading ambiguity resolution (CAR) method of fixing extra-wide-lane (EWL), wide-lane (WL) and narrow-lane (NL) ambiguities sequentially for multi-frequency AR instead of fixing full set of ambiguities simultaneously employed up to version 1.4.

The EWL ambiguities are derived by combining the ambiguities of the 2nd and 3rd frequencies, as well as those of the 2nd and 4th frequencies. The WL ambiguities are derived by combining the ambiguities of the 1st and 2nd frequencies. The NL ambiguities are ambiguities of 1st frequency such as L1, E1 and B1I. Integer ambiguity searches of EWL and WL are performed by simple rounding, while that for NL is performed by LAMBDA method.

The main functions are as follows.

function	Description
ppp_amb_ILS()	Ambiguity resolution by ILS (integer-least-square). Generate single-difference between satellites, search and fix extra-wide-lane, wide-lane and narrow-lane ambiguities and update EKF states.
gen_sat_sd()	Generate single-difference (SD) between satellites.
search_amb_ewl()	Search extra-wide-lane integer ambiguity.
search_amb_wl()	Search wide-lane integer ambiguity.
gen_sd_matrix_n1()	Generate SD-matrix of narrow-lane ambiguity.
search_amb_lambda()	Search integer ambiguity by LAMBDA.
update_states()	Update EKF states with integer ambiguity constraints.

(5) ppp_iono.c

There are functions for MADOCA-PPP ionospheric correction. In the MADOCALIB, the ionospheric corrections are added as a pseudo-observation to the EKF, thus constraining the states of ionospheric estimate to the ionospheric correction value. Also, before constrain by the ionospheric correction, the system bias between the delay by the STEC correction message and the delay by the estimated are removed described in section 6.5.2.2 of IS-QZSS-MDC.

The main functions are as follows.

function	Description
`const_iono_corr()`	Constraint to ionospheric correction. Estimate and removed system biases and constraint to external ionosphere correction.

Concluded